ATUL ABRAHAM

atulfabraham@gmail.com | (201) 344-9052 | linkedin.com/in/atulfranklin | github.com/AtulFranklin | www.atulabraham.site

SUMMARY

MS Computer Science student at Stevens Institute of Technology seeking Summer 2027 software engineering and fintech internships. Co-founded a trading-systems startup and built its F&O strategy testbed; merged an open-source PR fixing a 2x return overstatement in a Java backtester. Works in Python, Java and Dart with a focus on accuracy and anomaly detection.

EDUCATION

Stevens Institute of Technology, Hoboken, NJ **September 2026 - May 2028 (Expected)**
Master of Science in Computer Science

Vellore Institute of Technology, Chennai, India **July 2021 - July 2025**
Bachelor of Technology in Mechanical Engineering **GPA : 3.89/4.00**

PROJECTS

Pairs Trading Backtester: Open Source Contribution | *Java* | *Merged PR*
- Updated an open-source Kalman filter pairs trading backtester and added a CSV loader for 10 years of GLD/GDX prices.
- Fixed a max drawdown bug and a commission bug that overstated returns 2x (64.4% vs 32.9%), and added 34 JUnit tests to validate the risk and return analytics.
- Added performance metrics tracking (win rate, profit factor, Calmar ratio; 489 trades, 67.3% win rate); PR merged upstream after maintainer review.

Company Financial Analyzer | *Python, FastAPI, PostgreSQL, SEC EDGAR API* | *GitHub*
- Built a Python CLI data pipeline that pulls quarterly financials from the SEC EDGAR API and computes QoQ/YoY growth and operating margin for financial analysis.
- Handled derived Q4s, restatements and fiscal-quarter edge cases, with data validation and integrity checks, plus anomaly detection that flags bad data, backed by 78 unit tests.
- Added a FastAPI REST API with a PostgreSQL cache and SQL queries, covered by unit and integration tests.

Clean Spirit AI: Offline On-Device LLM Chat App | *Flutter, Dart, llama.cpp, LiteRT | Android* | *GitHub*
- Built an offline Android app running LLMs on-device (Gemma 4 via LiteRT on GPU, Qwen3 4B via llama.cpp).
- Engineered a dual-engine layer with automatic model selection, lazy loading and GPU-to-CPU fallback.

WORK EXPERIENCE

INVA Capital Markets, Remote **October 2025 - August 2026**
Co-Founder & Software Developer - Trading Systems | F&O Trading Automation and Strategy Testbed | Flutter, Dart, Firebase
- Designed and built a cross-platform F&O trading automation and strategy testbed for NSE/BSE with 5x intraday leverage, margin tracking and auto square-off; designed SQL databases to analyze 100+ simulated option trades.
- Added a 3-source market data pipeline (Yahoo, Stooq, NSE) and a Black-Scholes option chain with delta and PCR.
- Developed an automated scalping engine with RSI/volume confirmation, reversal detection and daily stop limits.
- Built predictive momentum models (a weighted 1W-6M score and a regime classifier needing 3 of 5 signals to agree) that screen 50 F&O stocks, with data visualization dashboards for scan results and P&L.

Whirlpool Corporation, Pune, India **March 2025 - September 2025**
Intern - Cost & Quality & Product Development
- Built a Google Apps Script (JavaScript) ETL and process automation tool that pulls SAP ERP component price CSVs for multiple product SKUs from a shared folder, cleans the data, groups it by SKU and formats it into standard cost spec sheets, cutting BOM upkeep time by over 50% through workflow optimization.
- Tool output became the cost spec source for managers, analysts and senior engineers, giving them automated reporting and data analysis for component cost optimization.
- Led 30+ corrective actions and validation tests and worked with 3 cross-functional teams on budget planning and Engineering Change Requests (ECRs).

Motherson Automotive Technologies & Engineering, Chennai, India **August 2023 - September 2023**
Trainee Intern - Quality Control (Exports)

SKILLS

Languages: Python, Java, Dart, SQL, JavaScript, HTML/CSS
Tools: Git, GitHub, Gradle, Android Studio,PostgreSQL, Google Apps Script, LLMs, Linux
Frameworks & Libraries: FastAPI, Flutter, Firebase (Auth), JUnit, unittest
Concepts: Data Structures, OOP, Unit Testing, API Integration, Database Design & Data Modeling, Query Optimization, ETL & Data Pipelines, Data Validation, Statistical Analysis, Risk Analytics, Backtesting, Options Pricing (Black-Scholes, Greeks), Agile